

```

%root_path = fileparts(mfilename("fullpath"));
%root_path = pwd;
scuts_path = 'e:\SHARE\Fe\Data\analysis\07_fit_with_multicut_allJ\sel_cuts';
Cutfitfile = fullfile(scuts_path,'GP_FFCutsFits.mat');

if ~exist('GP_FFCutsFits','var')
    ld = load(Cutfitfile);
    GP_FFCutsFits = ld.fit_src_struct;
end
fit_res_field = 'all_fit_par';

```

```

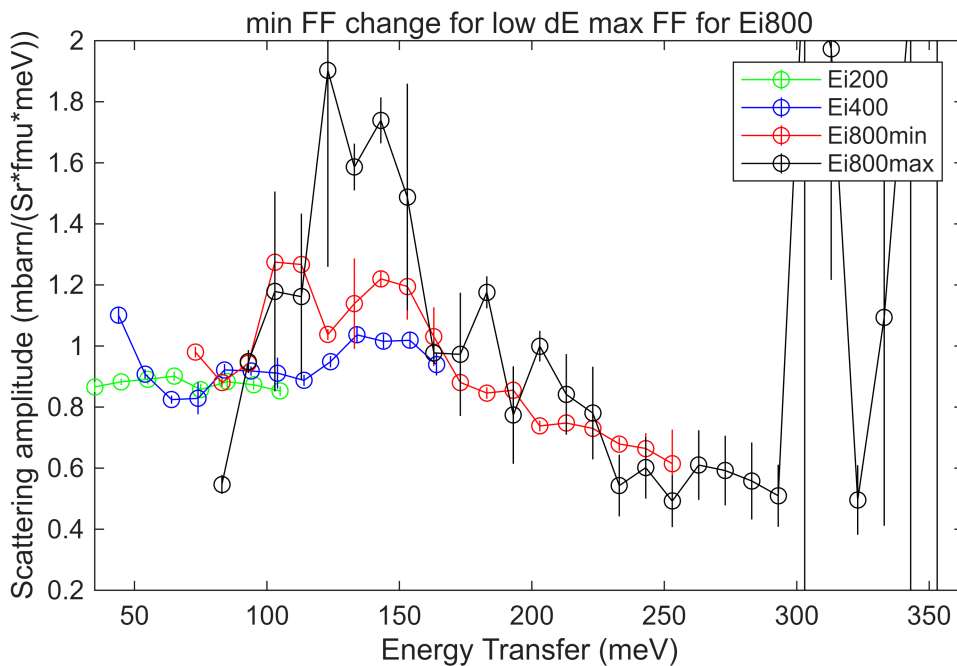
[S_op11Ei200,J_op11Ei200,Gam_op11Ei200,S_op11Ei200q] = extract_fit_par( ...
    GP_FFCutsFits.(['Ei200_off110GPmiff_',fit_res_field]));
[S_op11Ei400,J_op11Ei400,Gam_op11Ei400,S_op11Ei400q] = extract_fit_par( ...
    GP_FFCutsFits.(['Ei400_off110GPminFF_',fit_res_field]));
[S_op11Ei800,J_op11Ei800,Gam_op11Ei800,S_op11Ei800q] = extract_fit_par( ...
    GP_FFCutsFits.(['Ei800_off110GPminFF_',fit_res_field]));

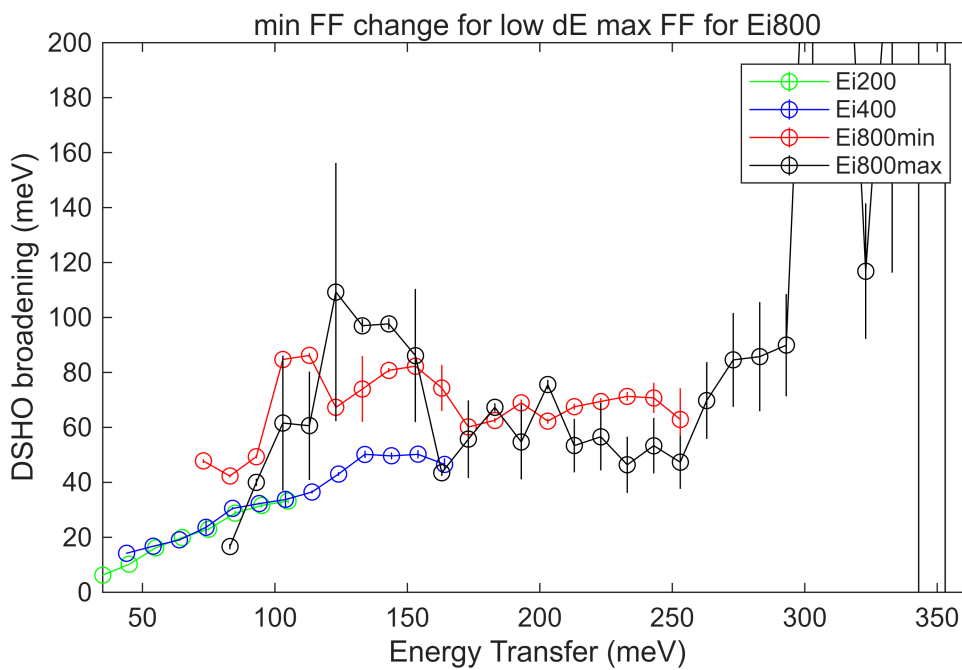
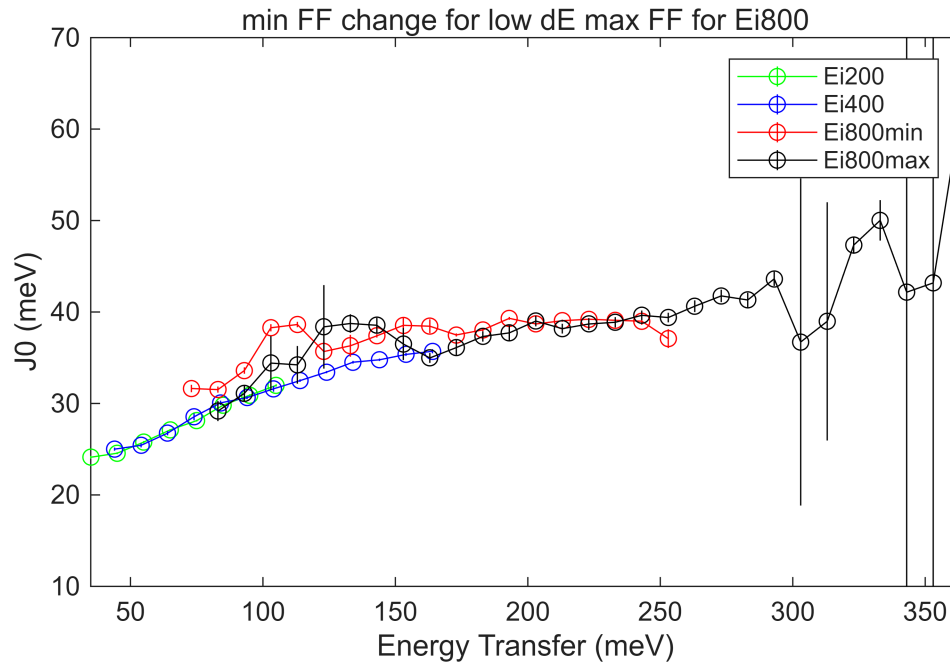
proj110minFF = GP_FFCutsFits.Ei200_off110GPmiff.data.proj;

[S_maEi800,J_maEi800,Gam_maEi800,S_maEi800q] = extract_fit_par( ...
    GP_FFCutsFits.(['Ei800_off200GPmaxFF_',fit_res_field]));
proj200maxFF = GP_FFCutsFits.Ei200_off200GPmaff.data.proj;

plot_fp('min FF change for low dE max FF for Ei800', ...
    [S_op11Ei200,S_op11Ei400,S_op11Ei800,S_maEi800], ...
    [J_op11Ei200,J_op11Ei400,J_op11Ei800,J_maEi800], ...
    [Gam_op11Ei200,Gam_op11Ei400,Gam_op11Ei800,Gam_maEi800]);

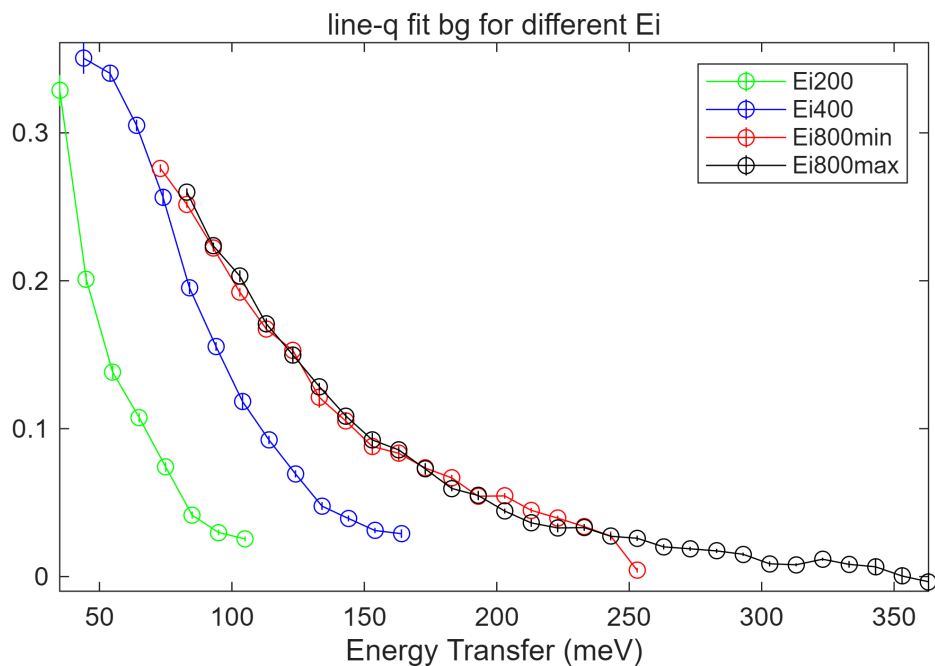
```



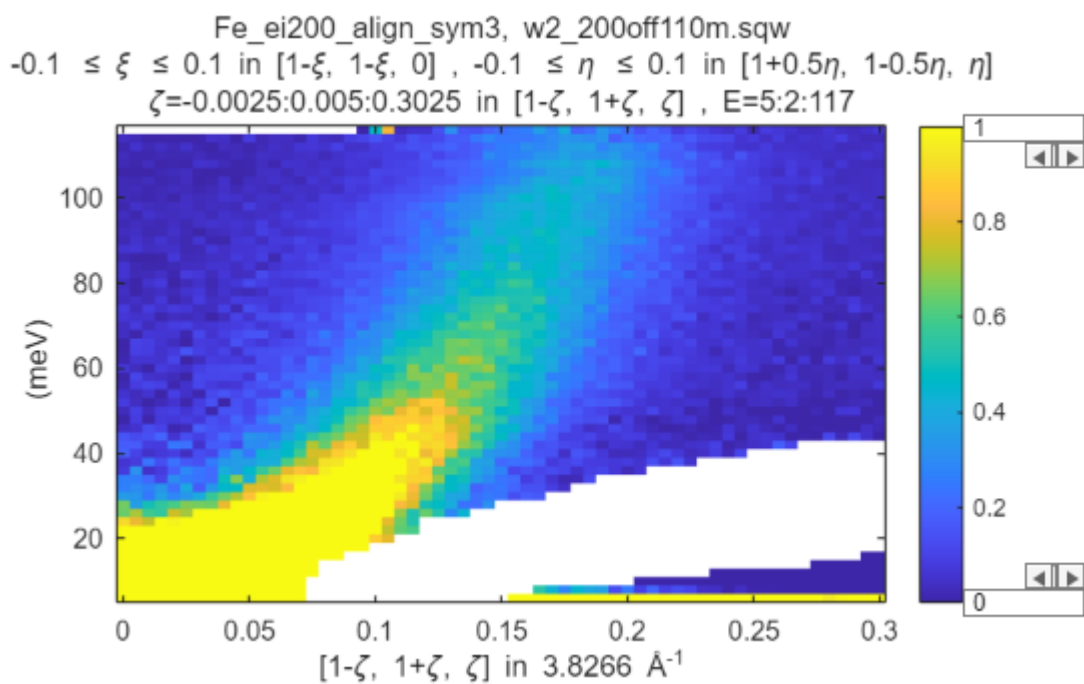


```
bg_dataEi200 = extract_bg_par(GP_FFCutsFits.
(['Ei200_off110GPmiff_',fit_res_field]),1);
bg_dataEi400 = extract_bg_par(GP_FFCutsFits.
(['Ei400_off110GPminFF_',fit_res_field]),1);
bg_dataEi800mi = extract_bg_par(GP_FFCutsFits.
(['Ei800_off110GPminFF_',fit_res_field]),1);
bg_dataEi800ma = extract_bg_par(GP_FFCutsFits.
(['Ei800_off200GPmaxFF_',fit_res_field]),1);

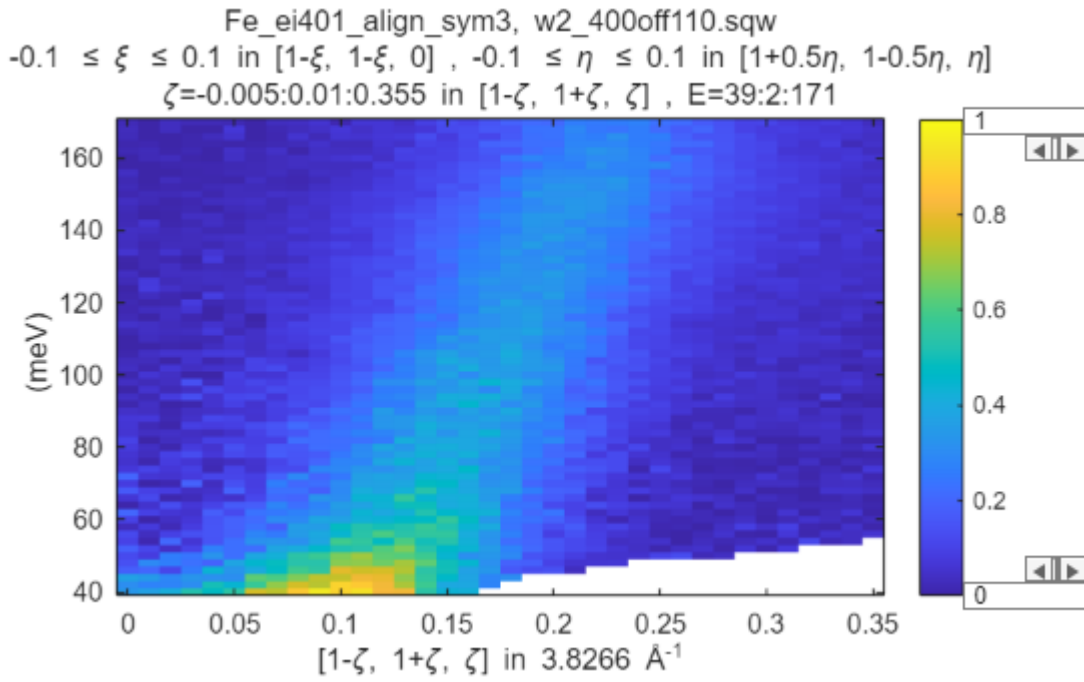
plot_fp('line-q fit bg for different Ei',[], ...
[bg_dataEi200,bg_dataEi400,bg_dataEi800mi,bg_dataEi800ma])
```



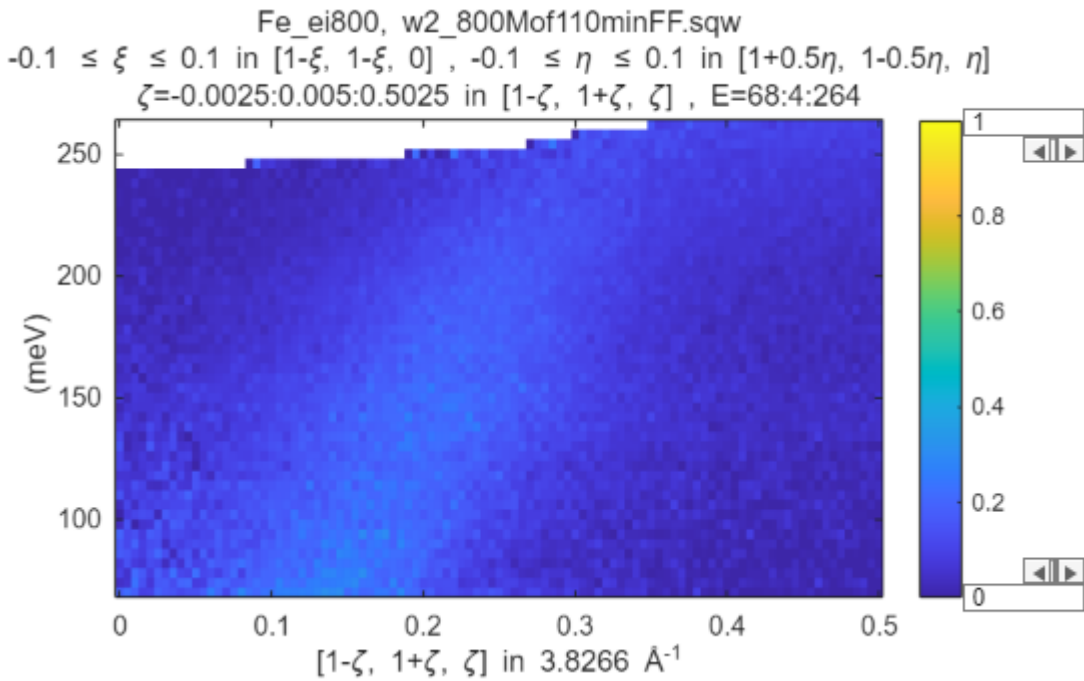
```
spp.Ei200_off110GPminNbg = GP_FFCutsFits.Ei200_off110GPmiff - ...
    sqw_eval(GP_FFCutsFits.Ei200_off110GPmiff,@linterp_bg,bg_dataEi200);
%plot(spp.Ei200_off110GPminNbg); lz 0 1; keep_figure
```



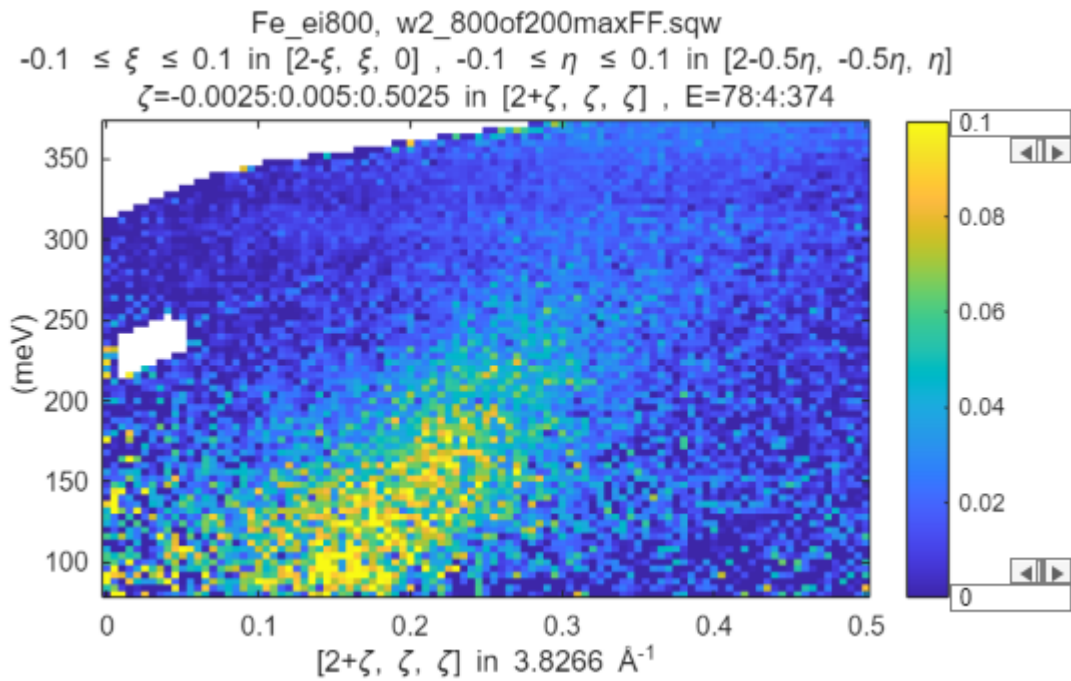
```
spp.Ei400_off110GPminNbg = GP_FFCutsFits.Ei400_off110GPminFF - ...
    sqw_eval(GP_FFCutsFits.Ei400_off110GPminFF,@linterp_bg,bg_dataEi400);
%plot(spp.Ei400_off110GPminNbg); lz 0 1; keep_figure
```



```
spp.Ei800_off110GPminNbg = GP_FFCutsFits.Ei800_off110GPminFF - ...
    sqw_eval(GP_FFCutsFits.Ei800_off110GPminFF,@linterp_bg,bg_dataEi800mi);
%plot(spp.Ei800_off110GPminNbg); lz 0 1; keep_figure
```



```
spp.Ei800_off200GPmaxNbg = GP_FFCutsFits.Ei800_off200GPmaxFF - ...
    sqw_eval(GP_FFCutsFits.Ei800_off200GPmaxFF,@linterp_bg,bg_dataEi800ma);
%plot(spp.Ei800_off200GPmaxNbg); lz 0 0.1; keep_figure
```

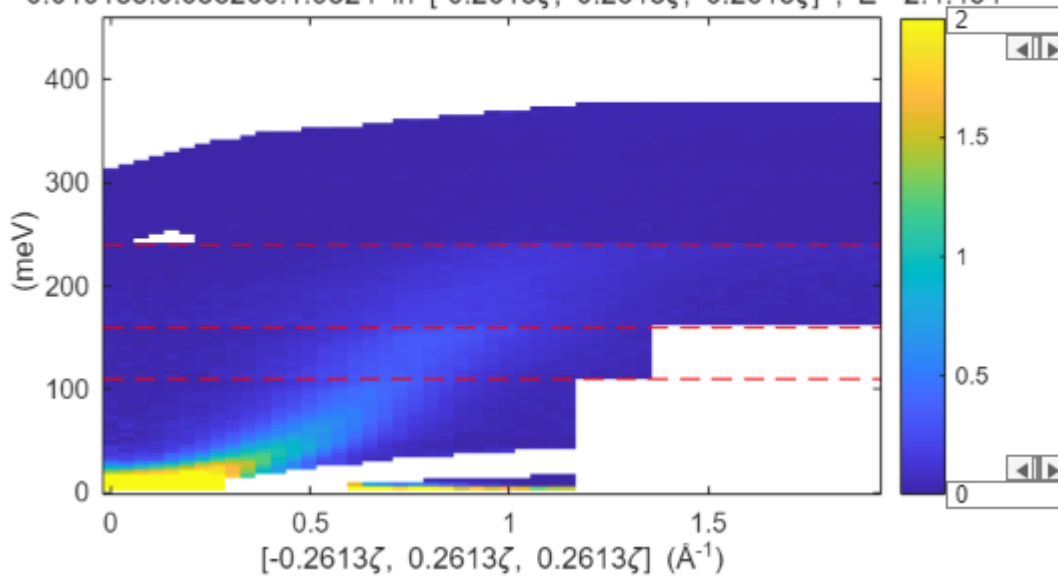


```
cuts_ranges = containers.Map();
cuts_ranges('Ei200_off110GPminNbg') = [0;110];
cuts_ranges('Ei400_off110GPminNbg') = [110;160];
cuts_ranges('Ei800_off110GPminNbg') = [160;240];
cuts_ranges('Ei800_off200GPmaxNbg') = [240;460];

[w2,fh] = build_disp_in_dir_with_ds(spp,cuts_ranges,0.01,0.5,false);
```

```
*** Finished 1of#4 chunks in 1.8701 sec performing sqw_op_bin_pixels
...
*** Completed sqw_op_bin_pixels using 4 pages in 4.351 sec.
*** Resulting object is backed by file:
```

Fe_ei200_align_sym3, w2_200off110m.sqw
 0.31244 in $[-0.3201\xi, -0.3201\xi, 0]$, $-0.27058 \leq \eta \leq 0.27058$ in $[0.1848\eta, -0.1848\eta, 0]$, $\zeta = -0.019133:0.038266:1.9324$ in $[-0.2613\zeta, 0.2613\zeta, 0.2613\zeta]$, $E = -2:4:454$



```
data_ranges = cuts_ranges.values;
data_ranges = [data_ranges{:}];
pan_data = struct('data_ranges',data_ranges,'combined_ds',w2);
%save('GP_0p50p50p5_data','pan_data');
fh.UserData = w2.data;
```

```
gamma=10;Seff0=0.7917;

J0=39; J1=35;
correct_ff = 1;
gap = 0; %
init_fg_param = [correct_ff,8,gamma,Seff0, 0, J0, J1, 0, 0, 0];

kk = tobyfit(spp.Ei800_off200GPmaxNbg);
kk = kk.set_fun(@sqw_iron);
hkl_proj = spp.Ei800_off200GPmaxNbg.data.proj;

%init_fg_params = [correct_ff,T,gamma,Seff, gap, J0, J1, J2, J3, J4];
free_sw_param = [0 0, 1, 1, 0, 1, 1, 0, 0, 0];

kk = kk.set_pin({init_fg_param,hkl_proj});
kk = kk.set_free(free_sw_param);

%kk = kk.set_bfun (@linear_bg1D); % set_bfun sets the background functions
%
%bg_param = init_bg_param(valid);
%kk = kk.set_bpin (bg_param); % initial background constant and gradient
%kk = kk.set_bfree (bfree);

kk = kk.set_options('list',2);
```

```
[fit_obj,fit_par] = kk.fit();
```

Warning: Dataset:59 points with zero, negative or undefined (infinite or NaN) error bars have been removed from fit, which is 0.7894

Beginning fit (max 20 iterations)-----

Starting point

Total time = 28.5491s Reduced Chi^2 = 12.3554

Free parameter values:

10	0.7917	39	35
----	--------	----	----

Iteration = 1

Total time = 161.9984s Reduced Chi^2 = 3.068 Levenberg-Marquardt = 0.1

Free parameter values:

20.37	0.4342	38.81	35
-------	--------	-------	----

Iteration = 2

Total time = 300.6549s Reduced Chi^2 = 1.9621 Levenberg-Marquardt = 0.01

Free parameter values:

51.92	0.543	38.4	35
-------	-------	------	----

Iteration = 3

Total time = 455.1789s Reduced Chi^2 = 1.6309 Levenberg-Marquardt = 0.001

Free parameter values:

102	0.7891	38.75	35
-----	--------	-------	----

Iteration = 4

Total time = 622.783s Reduced Chi^2 = 1.5804 Levenberg-Marquardt = 0.0001

Free parameter values:

133.3	0.9409	40.65	35
-------	--------	-------	----

Iteration = 5

Total time = 771.4976s Reduced Chi^2 = 1.5762 Levenberg-Marquardt = 1e-05

Free parameter values:

134.8	0.9413	41.69	35
-------	--------	-------	----

Iteration = 6

Total time = 930.868s Reduced Chi^2 = 1.575 Levenberg-Marquardt = 1e-06

Free parameter values:

132.8	0.9288	41.9	35
-------	--------	------	----

Fit converged:

Parameter values (free parameters with error estimates):

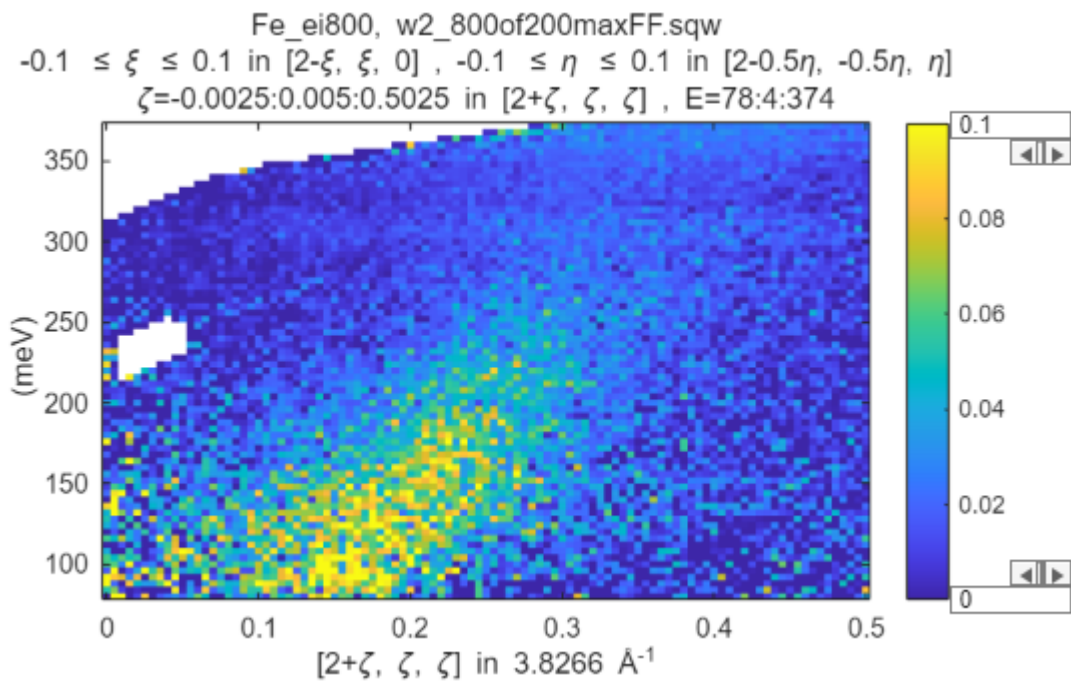
Foreground parameter values:

1	1		
2	8		
3	132.8	+/-	NaN
4	0.9288	+/-	NaN
5	0		
6	41.9	+/-	NaN
7	35	+/-	NaN
8	0		
9	0		
10	0		

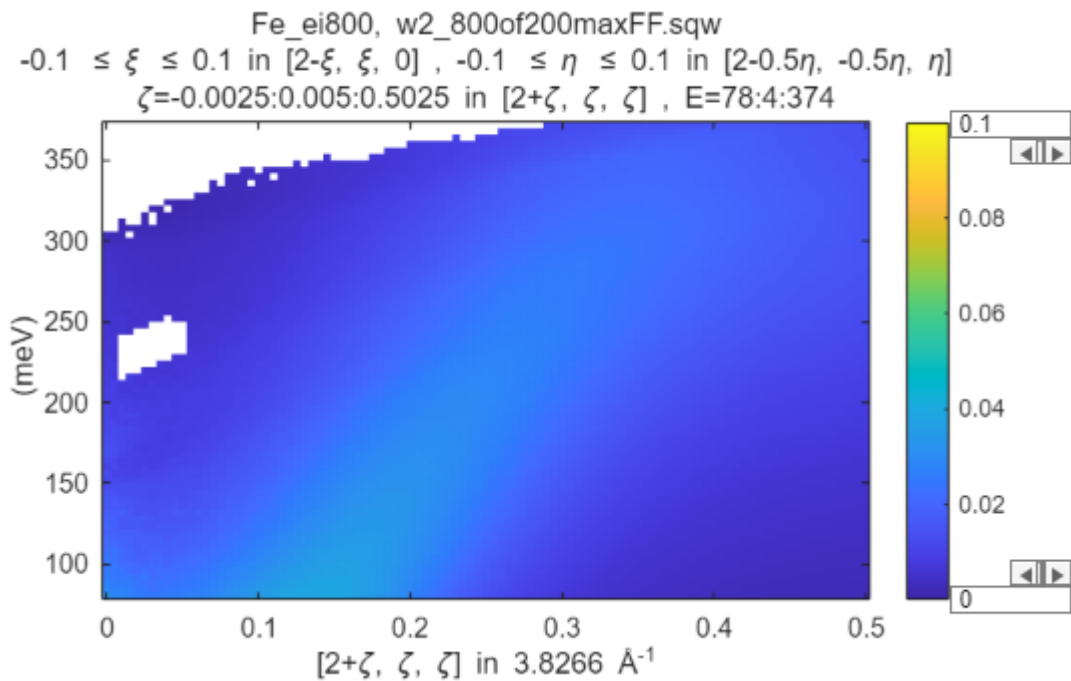
Covariance matrix for free parameters:

```
NaN NaN NaN NaN
NaN NaN NaN NaN
NaN NaN NaN NaN
NaN NaN NaN NaN
```

```
plot(spp.Ei800_off200GPmaxNbg); lz 0 0.1;keep_figure;
```



```
plot(fit_obj); lz 0 0.1;keep_figure;
```

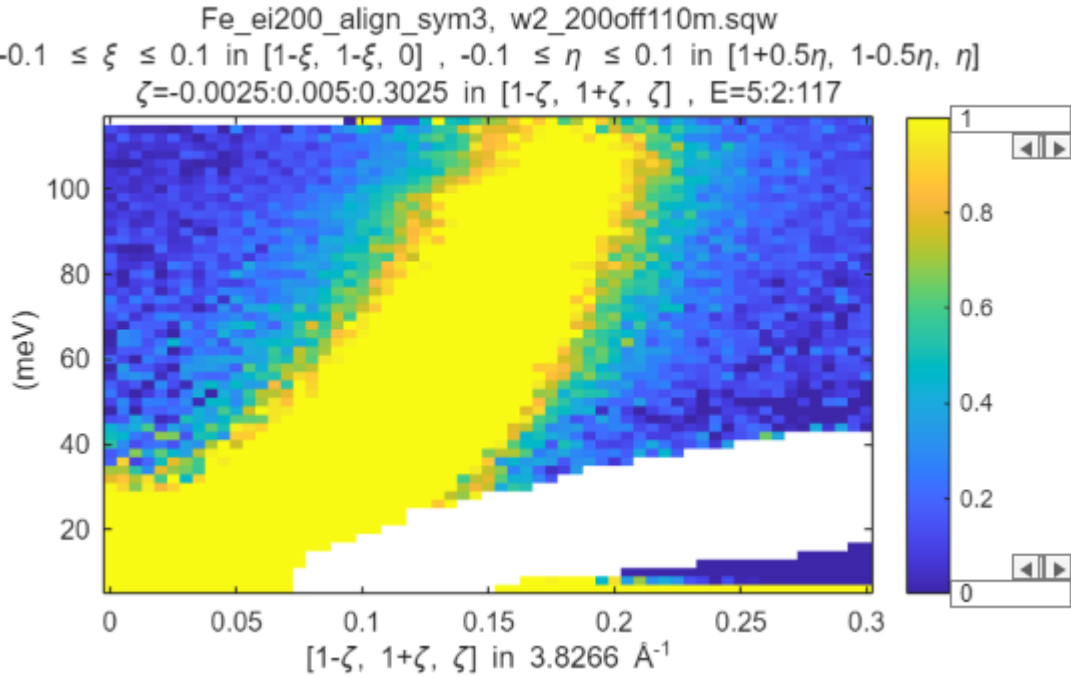


```
%[fit_obj,fit_par] = kk.simulate();
```

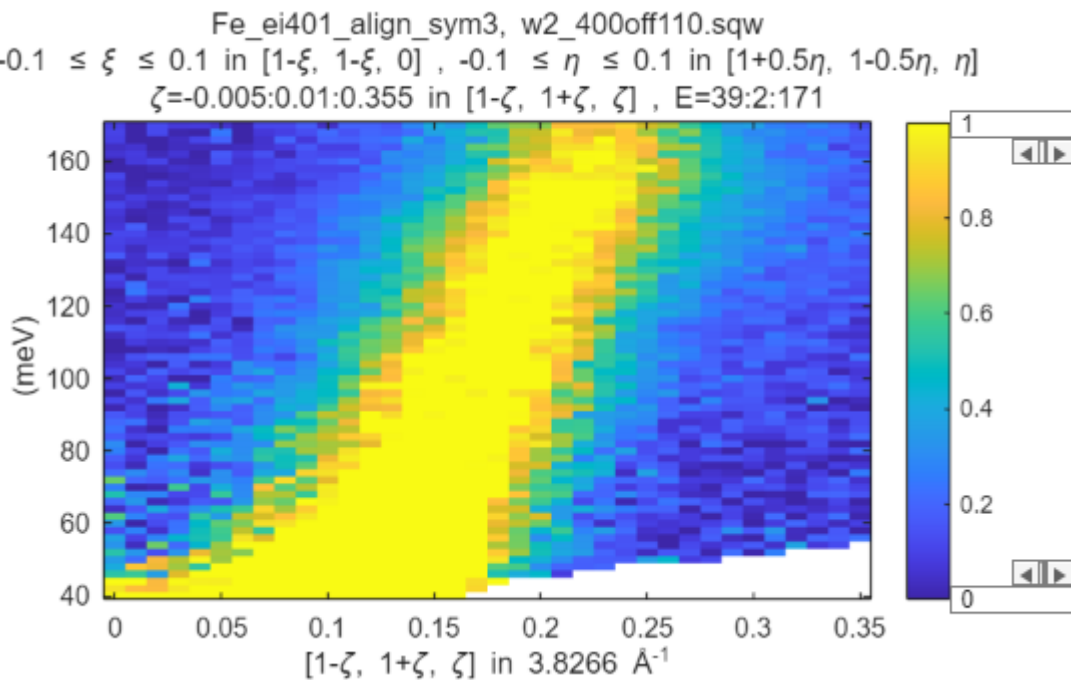
```
if ~exist('magi','var')
    magi = MagneticIons();
```

end

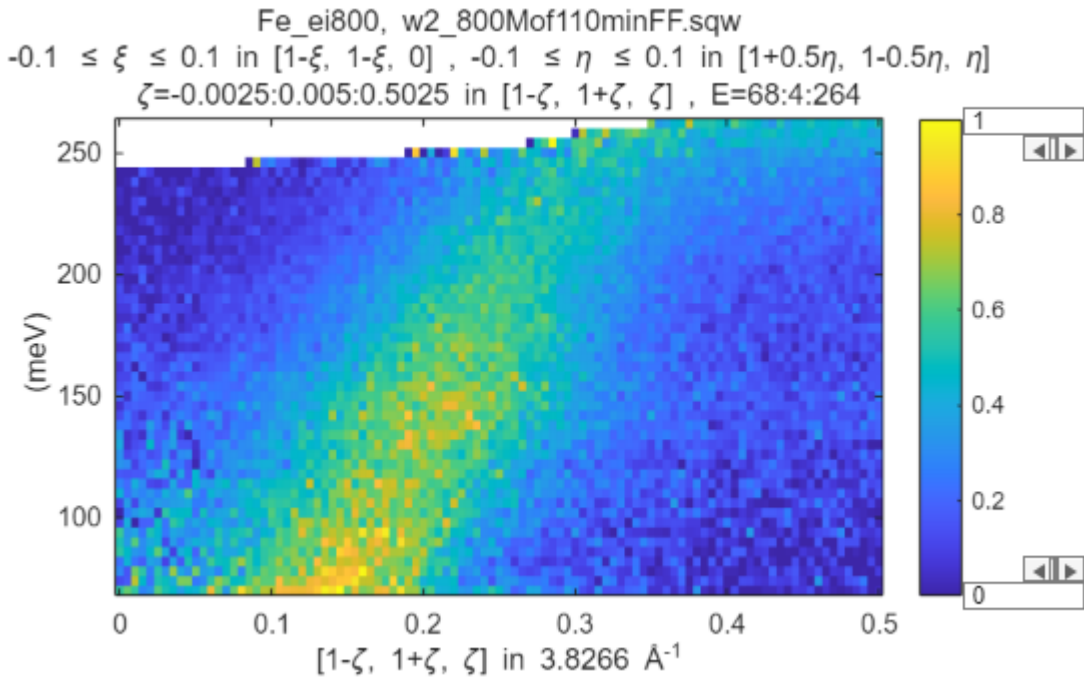
```
spp.Ei200_off110GPminNbg= magi.correct_mag_ff(spp.Ei200_off110GPminNbg);  
plot(spp.Ei200_off110GPminNbg); lz 0 1;keep_figure;
```



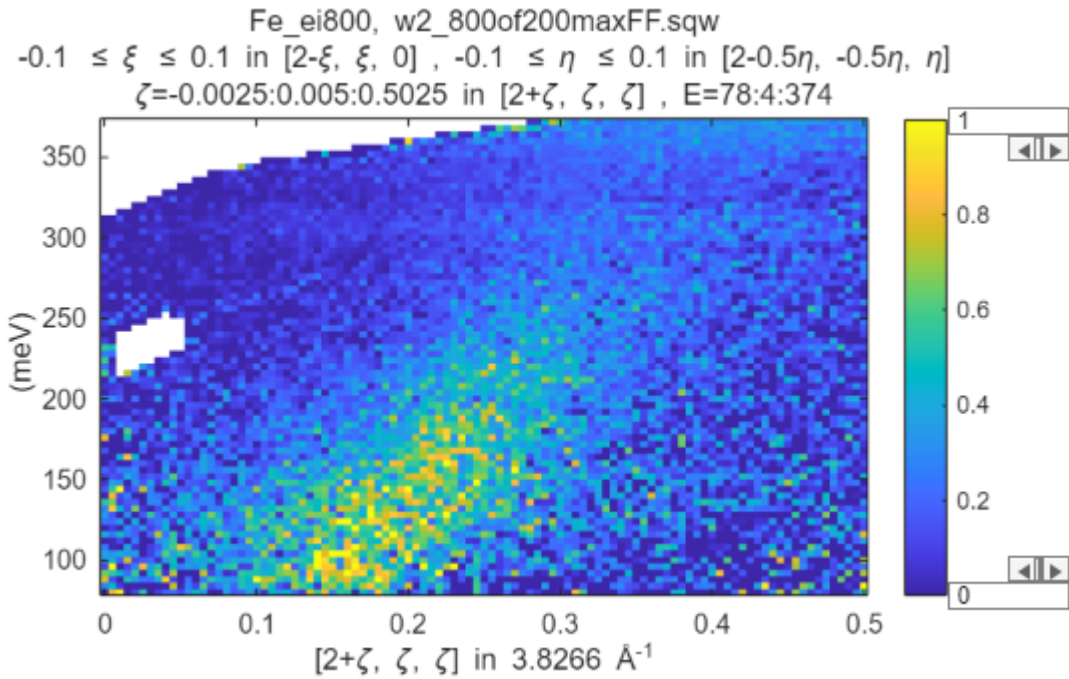
```
spp.Ei400_off110GPminNbg = magi.correct_mag_ff(spp.Ei400_off110GPminNbg);  
plot(spp.Ei400_off110GPminNbg); lz 0 1; keep_figure
```



```
spp.Ei800_off110GPminNbg = magi.correct_mag_ff(spp.Ei800_off110GPminNbg);  
plot(spp.Ei800_off110GPminNbg); lz 0 1; keep_figure
```



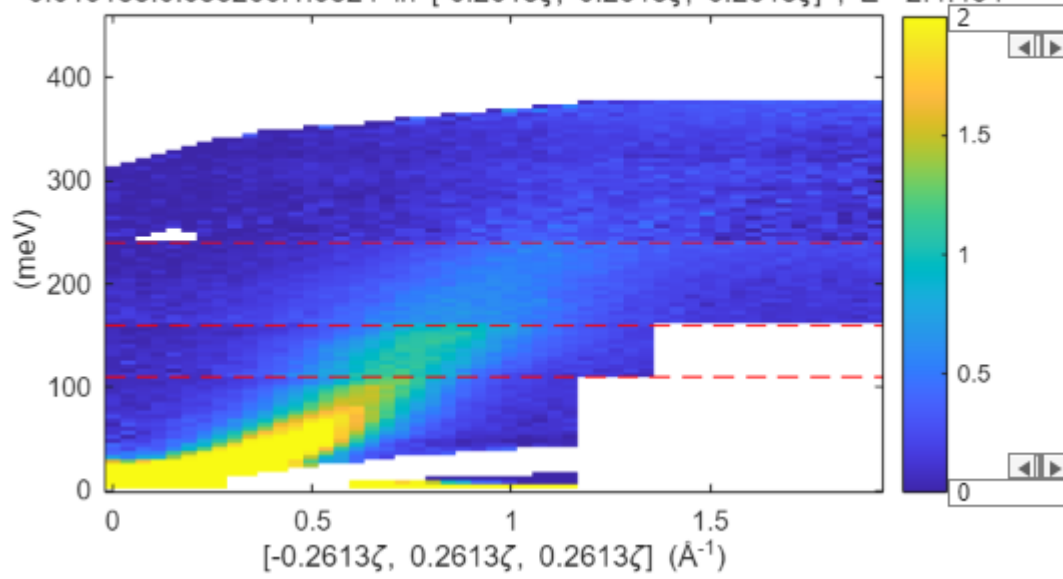
```
spp.Ei800_off200GPmaxNbg = magi.correct_mag_ff(spp.Ei800_off200GPmaxNbg);
plot(spp.Ei800_off200GPmaxNbg); lz 0 1; keep_figure
```



```
[w2,fh] = build_disp_in_dir_with_ds(spp,cuts_ranges,0.01,0.5,false);
```

```
*** Finished 1of#4 chunks in 1.7905 sec performing sqw_op_bin_pixels
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 0.31244 in $[-0.3201\xi, -0.3201\xi, 0]$, $-0.27058 \leq \eta \leq 0.27058$ in $[0.1848\eta, -0.1848\eta, 0]$,
 $\zeta = -0.019133:0.038266:1.9324$ in $[-0.2613\zeta, 0.2613\zeta, 0.2613\zeta]$, $E = -2:4:454$



```
data_ranges = cuts_ranges.values;
data_ranges = [data_ranges{:}];
pan_data = struct('data_ranges',data_ranges,'combined_ds',w2);
```